

CL-TAV: Closed-Loop Test–Analysis Verification for ARINC 615A Data Loading under Timing Uncertainty

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Abstract— ARINC 615A data loading must be judged under declared timers, file identities and measurement uncertainty. A fixed test suite that never uses residual ambiguity, or an analysis that never issues a later legal operation, can leave detection and isolation incomplete. This paper specifies Closed-Loop Test–Analysis Verification (CL-TAV): a charged session that predicts current observation classes, selects an exclusive TEST, Prep, Recover or Admit action, interprets timing with a shared uncertainty reference, and updates a single-fault candidate set only from a select-time snapshot. Unknown-effect ERROR is not implementation FAIL. Confirmed-not-sent leaves the known summary unchanged. The bound protocol model remains UPLOAD and INFORMATION; ARINC 645 identity is bound for triggered leaves, while integrity capabilities stay unestablished. Four comparison arms and three registered experiment families are defined. Confirmatory measurements are not yet obtained; improvement percentages are therefore not reported. The draft is a journalized method paper, not a solver implementation and not a certification argument.

Index Terms— Avionics, data loading, conformance testing, fault diagnosis, timing uncertainty, closed-loop verification.

I. Introduction

Airborne software loaders exchange INFORMATION and UPLOAD traffic under declared timers, file identities and operator actions [1]. A missed or late acknowledgement, a mismatched Load File, or an unresolved integrity check can be a protocol failure, a measurement artefact, or an unconfirmed injection. The engineering question is whether a later legal action can be chosen from residual ambiguity under a finite resource.

This paper studies Closed-Loop Test–Analysis Verification (CL-TAV). Three reader questions are used throughout:

- CL-RQ1 (loop consistency): do prediction, interpretation and candidate update stay mutually consistent under one uncertainty reference?
- CL-RQ2 (comparative effect): under the same budget and denominators, does the closed loop change

detection or isolation relative to Test-only, Analysis-only and fixed Test-then-Analysis arms?

- CL-RQ3 (uncertainty and stopping): how do interval timing, ERROR, Prep/Recover and exclusive stops affect those comparisons?

Two propositions are evaluated, not asserted as established results. C1 is uncertainty-aware interpretation, timing verdicts and compatible update (CL-RQ1). C2 is feeding residual ambiguity into the next exclusive action with explicit resource, Prep/Recover, ERROR and stop rules (CL-RQ2/3).

Requirements-based generation organises stimuli from obligations [2]. Conformance relations such as *ioco* generate tests from labelled transition systems [3]. FSM covering and mutant killing select discriminating sequences [4]–[7]. Section II compares those stated objects with CL-RQ1–CL-RQ3. The comparison is not a proof that those methods cannot be extended.

The evaluation uses four arms (CL-T, CL-A, CL-TA, CL-LOOP) and three registered families (DETECT, LOCATE, ABLATION). Confirmatory numbers are not available.

Placeholder PH-ABS-RESULT (EXPERIMENT-RESULT)

Abstract and conclusion improvement statements remain empty
until registered runs exist.

Data are not obtained. This box reserves layout. It is not a result.

Section II defines the objects. Section III separates CRS layers and the two machines. Section IV gives the loop, update and T5 mathematics. Section V freezes the comparison contract. Sections VI–VIII reserve results and limits. The separately compiled supplement type-sets wrappers ALG-CLTAV-02/03/04, core mechanisms 05/06/07, typed records and the A/P decision tables. Paper Algorithm numbers are display labels; stable ALG-IDs are unchanged.

II. Related Work and Problem Formulation

A. Selective comparison

Table I records the cited work’s stated object and its limited role in this paper. It makes no absence or laboratory-capability claim, and no paywalled method detail is used for a denial.

Tretmans develops an LTS implementation relation and test generation, not a charged isolation session [3]. Chow and Fujiwara *et al.* develop FSM covering sequences [4], [5]. Petrenko *et al.* develop constraint solving and mutant killing [6]. Jia and Harman survey mutation adequacy [7]. Yang *et al.* survey requirements-based generation [2]. Li *et al.* treat model-based testing of networked applications [8]. Alur and Dill supply timed-

TABLE I
Cited stated objects and their limited comparison role.

Work	Stated object / role here
Tretmans [3]	LTS implementation relation and test generation
Chow; Fujiwara <i>et al.</i> [4], [5]	FSM covering sequences
Petrenko <i>et al.</i> [6] Jia and Harman [7]	Constraint solving and mutant killing Mutation-adequacy survey
Yang <i>et al.</i> ; Li <i>et al.</i> [2], [8]	Requirements-based and networked-application testing
Alur and Dill [9]	Timed-automaton vocabulary used as timing language

automaton vocabulary used here as timing language, not as a deployed solver [9].

Placeholder PH-LIT-ACCESS (LITERATURE-ACCESS)

A later denial that needs a paywalled detail is not made until
AUTH-LIT obtains that text.

Data are not obtained. This box reserves layout. It is not a result.

B. Verification objects

The IUT is a 615A data-loading endpoint. The bound protocol model is UPLOAD and INFORMATION [1]. FIND, DOWNLOAD and conditional AFDX appear in the CRS; they are not implemented behaviour and not an experimental arm.

Candidates are $H_0 = \{h_{\text{normal}}\} \cup H_{\text{single}}$. An observable summary q is known or unknown. Observation classes are projected onto current H . Measurement uncertainty is a declared conservative outer set U in the GUM sense [10], [11], not an estimated physical error model. Resource is exactly one mode: BUDGET($c_{\min}, B$) or ROUNDS($K_{\max}$).

A 615A-triggered 645 bind covers CRC, check-value and naming leaves [12]. Integrity capabilities remain unestablished. Stop-645 names a remaining 645-dependent obligation, not a missing file.

III. System Engineering and Verification Architecture

Protocol CRS, tool requirements and experiment requirements stay in separate layers. Row count does not prove a method. A concrete owned trace is source unit SU-ARINC-615A-3-SECTION-6-3-2-SEQUENCE-CHART-A-E007-7726A816FFE7 $\rightarrow$ CRS-M1-00365/00366/00367 $\rightarrow$ LUR write-order constraint $\rightarrow$ planned Test/Analysis verification case $\rightarrow$ later tool module. The IDs are real bound-M1 identifiers. satisfy/verify in the SysML view are model

relations, not executed verification. No standard figure is reprinted.

Two machines must not be merged. M_{sess} selects from $S \subseteq A(q)$, charges once, updates H and returns a named stop. M_{prot} is the UPLOAD/INFORMATION black box. Execute issues a stimulus and obtains a raw execution record. Interpretation, not the network port, produces the compatible class I_z and hypothesis survivors.

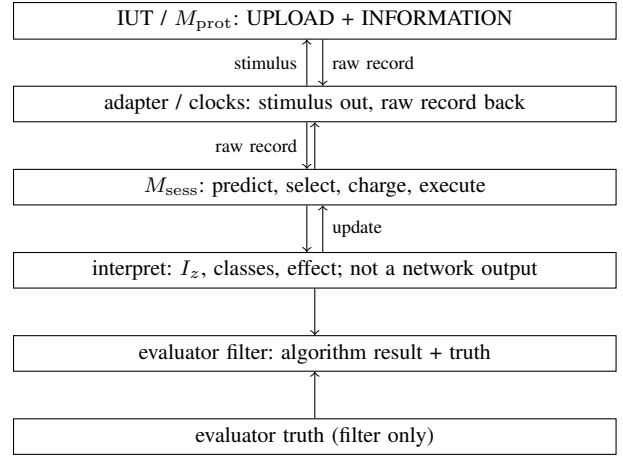


Fig. 1. Context derived from FIG-CL-TAV-01/07. Stimulus and raw records share ports. Truth enters only the evaluator filter, never select, predict, history or execute.

Expanded CRS leaves do not add FIND or DOWNLOAD transitions to M_{prot} . The development contract now binds eight algorithm interfaces and seven experiment interfaces to typed inputs, outputs, ownership, failure, resource, acceptance, and successor implementation locations. The first slice also preserves six CRC-bearing protocol obligations and their ARINC 645 parameter leaves. This closes their specification dependency only: the runtime integrity capability remains *not established* until implementation, project configuration, controlled-vector execution, and an independent gate record exist. Missing evidence therefore yields NOT-EVALUATED, not a protocol verdict.

IV. CL-TAV Method

A. Notation and assumptions

S0–S10 are logical duties, not line numbers. The decision index k , the billed charge count and the retry counter are distinct. SessionContext Γ is the unique session state. HistoryHandle η carries whole-history compatibility. The select-time snapshot ξ freezes actionId, actionKind, q_{used} , history version, classes used at select, and uncertainty reference U . A later confirmed successor summary is a different field. Effect classes are CONFIRMED-NOT-SENT and UNKNOWN-EFFECT. PredictionGapError is TEST-scoped.

Algorithm 1: CL-TAV top-level session (paper
Alg. 1 = ALG-CLTAV-01)

Require: $H_0 = \{h_{\text{normal}}\} \cup H_{\text{single}}$; Γ ; η ; one mode BUDGET($c_{\min}$, B) xor ROUNDS($K_{\max}$); L ; U

Ensure: stop class, final H , η , trace; singleton h_{normal} is not protocol PASS

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1  $(\Gamma, \eta, H) \leftarrow$ 
  INITIALIZE( $H_0, L, U, \text{resourcePolicy}$ );
2 while true do
3    $stop \leftarrow$  IF-RESOURCE-STOP( $\Gamma, H, \eta$ ) with
    IF-EQUIV( $\Gamma, H, \eta$ );
4   if  $stop \neq \text{undecided}$  then return
    FINISH( $stop, \Gamma, \eta, H$ );
5    $pred \leftarrow$  PREDICTCURRENT( $\Gamma, \eta, H, U, L$ );
6    $dec \leftarrow$  SELECTANDADMIT( $\Gamma, \eta, H, L, pred$ );
7   if  $dec.\text{kind} \in \{\text{EXIT}, \text{SPEC-ERROR}\}$  then
    return FINISH( $dec.\text{reason}, \Gamma, \eta, H$ );
8    $\xi \leftarrow$  SELECTSNAPSHOT( $dec; \Gamma, \eta, U$ );
9    $\xi = (\text{id}, \text{kind}, q_{\text{used}}, v_{\eta}, C, U)$ ;
10   $\text{kind} = \xi.\text{actionKind}$ ;
     $C = \xi.\text{classesUsedAtSelect}$ ;
11   $\Gamma \leftarrow$  CHARGEONCE( $\Gamma, \xi$ );  $er \leftarrow$ 
    IF-EXECUTE-RECORD( $\Gamma, \xi$ );
12   $z \leftarrow$  INTERPRETOUTCOME( $er, \xi, \Gamma, \eta, U$ );
13   $\delta \leftarrow$  RESOLVEOUTCOME( $z, \xi, \Gamma, \eta$ );
14   $\Gamma \leftarrow \delta.\Gamma$ ;  $\eta \leftarrow \delta.\eta$ ;  $z \leftarrow \delta.\text{outcome}$ ;
15  if  $\delta.\text{control} = \text{STOP}$  then return
    FINISH( $\delta.\text{reason}, \Gamma, \eta, H$ );
16  if  $\delta.\text{control} = \text{RETRY}$  then continue to S1
    with adopted  $\Gamma, \eta$ ;
17  if  $z.I_z$  valid or  $z.\text{summaryConfirmed}$  then
18     $upd \leftarrow$ 
      COMMITCOMPATIBLEUPDATE( $\xi, z; \chi$ );
19     $(\text{status}, \eta, H, \Gamma) \leftarrow upd$ ;
20    if  $\text{status} = \text{SPEC-ERROR}$  then return
      Stop-SpecError;
21  end
22   $stop \leftarrow$  IF-RESOURCE-STOP( $\Gamma, H, \eta$ ) with
    IF-EQUIV( $\Gamma, H, \eta$ );
23  if  $stop \neq \text{undecided}$  then return
    FINISH( $stop, \Gamma, \eta, H$ );
24 end

```

B. Overall closed loop

Algorithm 1 is a typed abstraction of ALG-CLTAV-01. Wrappers PREDICTCURRENT, INTERPRETOUTCOME and RESOLVEOUTCOME are ALG-CLTAV-02/03/04 in the supplement. S9 calls COMMITCOMPATIBLEUPDATE (ALG-CLTAV-07), with $\chi = (\Gamma, \eta, H)$, which validates the history version, applies IF-HIST-UPDATE from ξ , returns $(\text{status}, \eta, H, \Gamma)$ and commits a confirmed successor summary. A bare IF-HIST-UPDATE call is not that wrapper. Resolution is branched on $\delta.\text{control}$ after adopting $\delta.\Gamma$, $\delta.\eta$ and $\delta.\text{outcome}$.

C. Interpretation, compatibility and selection

Prediction and interpretation share U . Let $O(h, t, q)$ be the conservative outer set of feasible observations of hypothesis h , and let Z_k be the measured feasible-observation set after timing, ownership and pairing checks. The interpretation/whole-history bridge induces the compatible hypothesis class C_k (the accepted interface field $z.I_z$); C_k is not a measurement set. A currently valid class is the nonempty projection $H \cap C_k$. Compatible update on a valid class is

$$H_{k+1} = H_k \cap C_k, \quad H_{k+1} \subseteq H_k. \quad (1)$$

ERROR and identity Prep/Recover leave H unchanged. The one-step score on currently valid nonempty classes of an affordable distinguishing TEST t is

$$s(t) = \max_{C \in \mathcal{C}_t} |C|. \quad (2)$$

Empty S is never arg min. Timing uses T5 [11]: $I_{\text{obs}} \subseteq I_r$ is PASS; $I_{\text{obs}} \cap I_r = \emptyset$ is FAIL; otherwise INCONCLUSIVE. An ambiguous pair is ERROR, not IUT FAIL.

D. Bounded reference-kernel contract

The first implementation target is a finite explicit-path kernel, not a general solver. Each frontier entry contains a stable hypothesis ID, control state, exact rational clock constraint, finite typed store, path length, whole-history provenance, and a known/conservative-unknown status. It accepts only the registered Boolean, state-equality, typed-field-equality and rational interval guard AST, with finite assignment and zero-clock-reset updates. Propagation validates the select-time snapshot, expands enabled transitions, intersects observation and timing constraints, and retains every feasible successor. Entries merge only when hypothesis, control state, clock constraint, store and history provenance are identical.

Configured bounds cover hypotheses, actions, observation classes, path length and per-hypothesis frontier size. Unsupported syntax or an exhausted path or frontier bound returns the interface's named GAP, SPEC-ERROR, or conservative unknown result without deleting a hypothesis. This contract maps the internal result separately to all eight established interfaces; it does not claim completeness, polynomial complexity, or deployed code.

E. Conditional properties

Monotonicity. Equation (1) gives $H_{k+1} \subseteq H_k$. Without a valid class there is no add or delete. An in-domain singleton is not protocol PASS.

Retention. Assume (R1) $h^* \in H_0$; (R2) one state path is consistent with the executed history; (R3) $O(h^*, t_k, q_k)$ is a conservative outer set; (R4) the true observation belongs to both $O(h^*, t_k, q_k)$ and Z_k ; and (R5) ownership and pairing are valid. The declared interpretation and whole-history compatibility contract therefore places h^* in C_k . Thus $h^* \in H_k \cap C_k$, and induction retains it in

H_{k+1} . Out-of-domain faults may still mimic an in-domain survivor.

One-step minimax. On nonempty finite T_{cur} , t^* is a least element of $(s(t), \text{cost}(t), \text{id})$. This is not a global policy.

Termination. Every charged TEST, Prep, Recover and ERROR spends the selected mode, hence at most $\lfloor B/c_{\min} \rfloor$ or $K_{\max}$ billed executions. Wall-clock waits still need their own timeouts; those timeouts are not claimed implemented.

Exclusive named stops are Empty, Singleton, 645, Equivalent, Budget, Stop-Error and Stop-NoDistinguisher. Stop-Error is not Stop-Budget. Admit reasons A2–A5 and post-update P1–P5 are tabulated in the supplement. CONFIRMED-NOT-SENT does not assign UNKNOWN. UNKNOWN-EFFECT must not be read back as the previous KNOWN summary. Unconfirmed Recover with $R_{\max} = 1$ and remaining budget is Stop-Error. An executed empty H is Stop-Empty. Kernels remain unimplemented interfaces.

V. Experimental Setup and Evaluation Protocol

The comparison contract is the existing experiment plan. Four arms share the same information pool, action pool and fault domain. Direct numeric comparison is allowed only on the same denominator and the same charged projection of $c = (c_{\text{acq}}, c_{\text{comp}}, c_{\text{prep}}, c_{\text{retry}})$. Seven controlled interfaces define scene registration, injection, independent truth, causal collection, arm execution, validity filtering and evaluation. Historical captures use only the offline evidence line and are not runnable scenes or confirmatory truth. The controlled line still depends on an interactive IUT, confirmed injection, reset evidence, qualified clocks and a truth generator whose shared components are disclosed.

- CL-T: predetermined tests; projection c_{acq} for tests that run.
- CL-A: analysis on pre-allowed records; projection c_{comp} only; no invented sampling cost.
- CL-TA: frozen collect then analyse; c_{acq} and c_{comp} reported separately.
- CL-LOOP: the CL-TAV session; charge the components that actually run.

Truth is evaluator-only. It never enters IF-SELECT-ADMIT, IF-PRED-OBS, IF-HIST-UPDATE or IF-EXECUTE-RECORD. Report four rates on every arm: attempt coverage (all registered attempts); refusal/no-answer among attempts; conditional error on the answered subset; miss on the detection denominator after invalid and equivalent faults are removed. Abstention is first-class refusal and is excluded from PASS/FAIL.

DETECT compares miss and refusal under the same detection budget and the T5 oracle; FIND abort does not waive the 2 s / 3 s clocks. LOCATE compares surviving $|H|$ and true-fault exclusion on confirmed in-domain

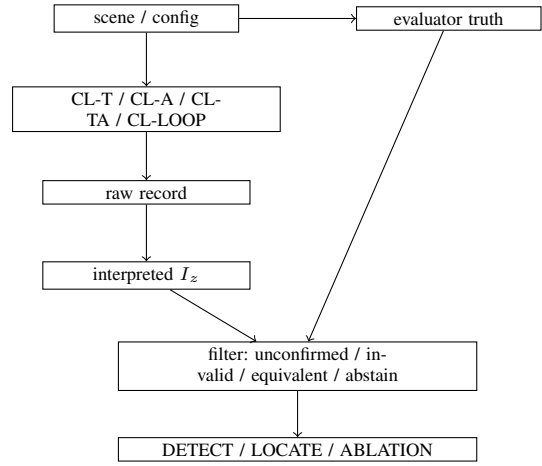


Fig. 2. Experiment architecture from FIG-CL-TAV-09. Truth meets algorithm output only at the evaluator filter.

TABLE II

Reserved ablation cells. CL-A is computation-only; N/A cells are not silent omissions. FIND-abort waiver is a forbidden control.

Cell	CL-T	CL-A	CL-TA	CL-LOOP
Charged projection	c_{acq}	c_{comp}	split	actual
No-feedback contrast	yes	N/A	yes	reference
Prep/Recover uncharged	if run	N/A	if run	contrast
Interval vs point stamp	—	—	—	contrast
FIND-abort waiver	forbidden	forbidden	forbidden	forbidden

injections. ABLATION uses the registered factors: no feedback (CL-T or CL-TA); Prep/Recover uncharged; interval versus point-timestamp on the same traces with clocks fixed; and FIND-abort-as-waiver as a forbidden negative control, not an allowed algorithm. Sample size remains a named gap.

Placeholder PH-STAT-N (PUBLICATION-DECISION)

Numeric n , repeats and interval method freeze at confirmatory registration, not in this draft.

Data are not obtained. This box reserves layout. It is not a result.

VI. Results

No effect direction, percentage or superiority is reported.

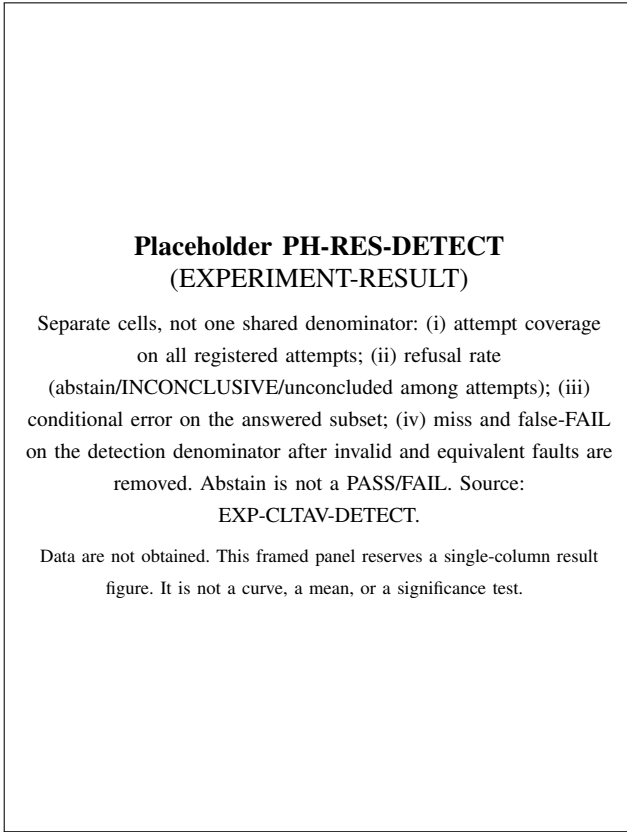


Fig. 3. Reserved detection panel. Not a result.

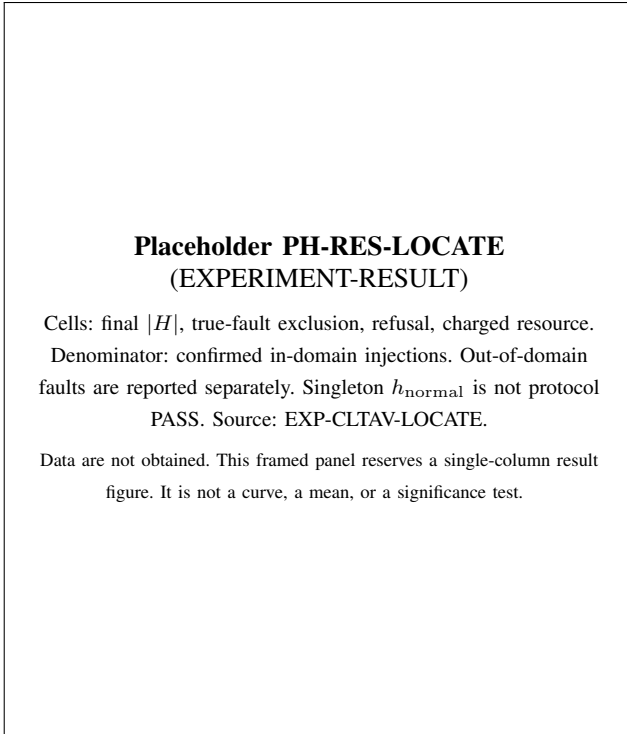
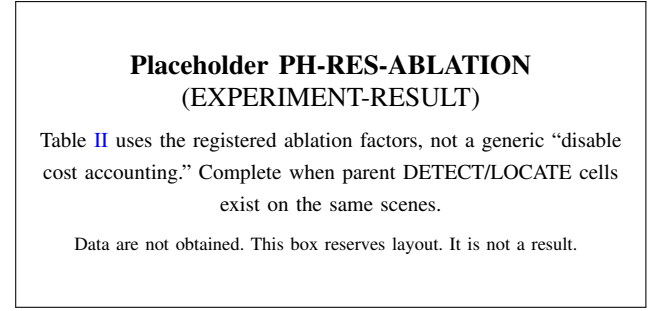


Fig. 4. Reserved localization panel. Not a result.

- A. CL-RQ2: verdict reliability
- B. CL-RQ2: isolation
- C. CL-RQ3: cost and sensitivity

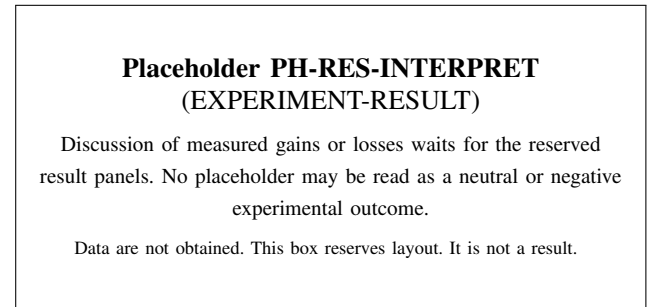


VII. Discussion and Limitations

The first-version domain is a declared single-fault H_0 , one-step selection, a bound UPLOAD/INFORMATION model, and notation-only architecture views. Out-of-domain faults may remain compatible with an in-domain survivor. A remaining $\{h_{\text{normal}}\}$ is not a protocol PASS. Obtaining ARINC 645 and binding triggered leaves does not close the four integrity capabilities.

Engineering work that is not done includes the core estimators, an executable verification engine, Project Configuration and confirmatory runs. Those absences are unfinished work, not method impossibility.

The 23 inspected captures are a selected exploratory corpus with unknown topology, clock qualification, device configuration, software version and independent root-cause truth. Repeated sessions and shared parser, model or clock components induce dependence and common-error risk. They can exercise intake and reconstruction robustness, but cannot support active counterfactual comparison among the four experimental arms. The bounded reference kernel also abstains rather than treating a frontier limit as incompatibility.



Transfer beyond Ethernet laboratory loaders, and any certification argument, remain open (historical RQ6).

VIII. Conclusion

CL-TAV specifies a charged observation–update–select session for ARINC 615A data loading under declared uncertainty and one resource mode. The main-text loop keeps interpretation, compatible update and

exclusive stops consistent with the accepted algorithm package. Experimental answers to CL-RQ2 and CL-RQ3 are not available.

**Placeholder PH-CONC-RESULT
(EXPERIMENT-RESULT)**

The concluding experimental sentence stays empty until
PH-RES-DETECT, PH-RES-LOCATE and PH-RES-ABLATION
are cleared by registered runs.

Data are not obtained. This box reserves layout. It is not a result.

This draft is not submission-ready, not an implemented solver, and not capability establishment.

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